

BOOTSTRAPPING IN SURVIVAL ANALYSIS

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Key words: Bootstrap; Kaplan-Meier estimator; regression model; weak convergence.

Summary: In this survey we consider the model of right random censorship in which the quantity of primary interest is the lifetime distribution function F . Kaplan and Meier (1958) derived a nonparametric maximum likelihood estimator F_n for F , which is a natural generalization of the empirical distribution function in the complete data case. It was Efron (1981) who first proposed procedures for the construction of bootstrap observations, which then produce a bootstrapped version F_n^* of F_n . We discuss the properties of F_n and F_n^* which lead to the important consistency result: almost surely, as $n \rightarrow \infty$, the process $n^{1/2}(F_n^* - F_n)$ converges weakly to the same Gaussian limit as the original Kaplan-Meier process $n^{1/2}(F_n - F)$. We also deal with the analogous properties for the corresponding quantile processes. Moreover, we consider the question of accuracy of the bootstrap approximation. Finally, we look into the situation of regression models which study the effect of covariates on the distributions of the lifetimes. We consider bootstrap procedures for the regression parameter in Cox's proportional hazards model and for Beran's distribution function estimator in a fixed design nonparametric regression model.

1. Introduction

In survival analysis, the random variable of interest Y is called the lifetime or survival time and typically represents the time to death of a patient (in medical trials) or the time to failure of a certain component (in industrial trials). A

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special difficulty in the analysis of such data is that some of the lifetimes may not be entirely observed. For example, the occurrence of a death may be censored by some other event that happens earlier in time, and we only know that the lifetime exceeds a certain value. In a medical study, for instance, it might happen that for some patients the true survival time is censored because the patient is lost to follow-up or drops out. The type of censoring considered here is random right censoring, but it should be noted that many other types can occur: left censoring, interval censoring, truncation, ... Some of the standard textbooks on survival analysis are Gill (1980), Miller (1981), Fleming and Harrington (1991), Andersen, Borgan, Gill and Keiding (1993).

2. The right random censorship model

A way to describe the above mechanism of right censoring is the following. Let $Y_1, \dots, Y_n$ be independent and identically distributed (i.i.d.) random variables (r.v.), called survival times, which are at risk of being censored from the right. That is, along with the $Y_1, \dots, Y_n$ we also have another sequence $C_1, \dots, C_n$ of i.i.d. r.v., called censoring times, and we only observe the pairs $(T_1, \delta_1), \dots, (T_n, \delta_n)$, where for $i = 1, \dots, n$, $T_i = \min(Y_i, C_i)$ and $\delta_i = I(Y_i \leq C_i)$. Here $I(A)$ denotes the indicator function of the event A . In this way, δ_i takes the value 1 if the i -th observation is uncensored ($Y_i \leq C_i$) and the value of 0 if the i -th observation is censored. (As is mostly the case in survival analysis, we will assume that the Y 's and C 's are non-negative, although this is not essential for the model). The distribution functions of the Y 's and the C 's are denoted by F and G respectively.

The model of random right censorship makes the assumption that Y_i and C_i are independent for each i . This independence assumption entails that $T_1, \dots, T_n$ are i.i.d. with distribution function H , satisfying $1 - H = (1 - F)(1 - G)$, and that $\delta_1, \dots, \delta_n$ are i.i.d. Bernoulli distributed r.v. with $\gamma = E(\delta_1) = P(\delta_1 = 1) = \int_0^\infty (1 - G(s-))dF(s)$.

3. The product-limit estimator of Kaplan and Meier

The quantity of primary interest in the model of random right censorship is the lifetime distribution function F of the random variable Y . Kaplan and Meier (1958) derived a nonparametric maximum likelihood estimator for F , based on the observations $(T_1, \delta_1), \dots, (T_n, \delta_n)$. This product-limit estimator is the natural generalization of the empirical distribution function when the data are complete (i.e. $T_i = Y_i$ for all i).

An interesting way of introducing this estimator (different from the treatment of Kaplan and Meier) is given below. It is based on relating F to quantities that can be estimated from the observations.

Introduce the following notation: for any right continuous function A : $\Delta A(t) = A(t) - A(t-)$ and $A^c(t) = A(t) - \sum_{s \leq t} \Delta A(s)$.

The cumulative hazard function corresponding to F is defined as

$$\Lambda(t) = \int_0^t \frac{1}{1 - F(s-)} dF(s).$$

We have the following relation (see Gill, 1980; Gill and Johansen, 1990; Andersen, Borgan, Gill and Keiding, 1993):

$$1 - F(t) = \exp(-\Lambda^c(t)) \prod_{s \leq t} (1 - \Delta\Lambda(s)). \quad (3.1)$$

Introduce the subdistribution function of the uncensored observations

$$H^u(t) = P(T_1 \leq t, \delta_1 = 1) = \int_0^t (1 - G(s-)) dF(s).$$

Then, we have

$$\Lambda(t) = \int_0^t \frac{1 - G(s-)}{(1 - F(s-))(1 - G(s-))} dF(s) = \int_0^t \frac{1}{1 - H(s-)} dH^u(s).$$

Since the quantities $H(t)$ and $H^u(t)$ are directly estimable from the data $(T_1, \delta_1), \dots, (T_n, \delta_n)$ by, respectively,

$$H_n(t) = \frac{1}{n} \sum_{i=1}^n I(T_i \leq t)$$

$$H_n^u(t) = \frac{1}{n} \sum_{i=1}^n I(T_i \leq t, \delta_i = 1)$$

we have the following estimator for $\Lambda(t)$:

$$\Lambda_n(t) = \int_0^t \frac{1}{1 - H_n(s-)} dH_n^u(s).$$

This is the Nelson-Aalen estimator for $\Lambda(t)$ and via relation (3.1), this gives the following estimator for $F(t)$:

$$1 - F_n(t) = \prod_{s \leq t} (1 - \Delta\Lambda_n(s)).$$

This estimator $F_n(t)$ is the product-limit or Kaplan-Meier estimator for $F(t)$. The function $\Lambda_n(t)$ has a jump only when $t = T_i$ with $\delta_i = 1$ and the size of the jump at T_i equals

$$\frac{\frac{1}{n} \# \{j : T_j = T_i\}}{1 - H_n(T_i -)} = \frac{\sum_{j=1}^n I(T_j = T_i)}{\sum_{j=1}^n I(T_j \geq T_i)} = \frac{m_i}{M_i}.$$

Hence, with these notations, the Kaplan-Meier estimator can be expressed as

$$1 - F_n(t) = \prod_{T_i \leq t} \left(1 - \frac{m_i}{M_i}\right)^{\delta_i}. \quad (3.2)$$

If all T_i are different, then each $m_i = 1$ and $M_i = n - \text{rank}(T_i) + 1$, and in this case

$$\begin{aligned} 1 - F_n(t) &= \prod_{T_i \leq t} \left(1 - \frac{1}{n - \text{rank}(T_i) + 1}\right)^{\delta_i} \\ &= \prod_{T_{(i)} \leq t} \left(1 - \frac{1}{n - i + 1}\right)^{\delta_{(i)}} = \prod_{T_{(i)} \leq t} \left(\frac{n - i}{n - i + 1}\right)^{\delta_{(i)}} \end{aligned} \quad (3.3)$$

where $T_{(1)} \leq T_{(2)} \leq \dots \leq T_{(n)}$ are the order statistics of $T_1, \dots, T_n$ and $\delta_{(1)}, \dots, \delta_{(n)}$ are the corresponding δ 's (i.e. $\delta_{(i)} = \delta_j$ if $T_{(i)} = T_j$).

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- (1) If there are ties among the $T_1, \dots, T_n$, then we make the following convention: in case of a tie between a censored and an uncensored, the uncensored

is ranked just before the censored. If at a value T_i , there are m_i uncensored observations and if $M_i = \#\{j : T_j \geq T_i\}$, then the contribution of these m_i uncensored observations to the product in formula (3.3) is given by

$$\frac{M_i - 1}{M_i} \cdot \frac{M_i - 2}{M_i - 1} \cdots \frac{M_i - m_i}{M_i - m_i + 1} = \frac{M_i - m_i}{M_i} = 1 - \frac{m_i}{M_i}$$

which corresponds to formula (3.2).

- (2) If there is no censoring (i.e. all $\delta_i = 1$), then it is easily seen that $F_n(t)$ reduces to the usual empirical distribution function.
- (3) If the last observation $T_{(n)}$ is censored, then the definition of $F_n(t)$ gives that it is strictly less than 1 and constant for $t \geq T_{(n)}$. There exist various proposals for redefining $F_n(t)$ for $t \geq T_{(n)}$.

4. Efron's bootstrap in the right random censorship model

It was Efron (1981) who first proposed bootstrap procedures for right randomly censored observations. His first proposal is to take independent resamples from the Kaplan-Meier estimators of the lifetimes and the censoring times, and to combine them by taking minima and indicators.

Efron's bootstrap procedure I:

- (1) Resample independently

$Y_1^*, \dots, Y_n^* \stackrel{i.i.d.}{\sim} F_n$ (Kaplan-Meier estimator for F)

$C_1^*, \dots, C_n^* \stackrel{i.i.d.}{\sim} G_n$ (Kaplan-Meier estimator for G)

- (2) Form $(T_1^*, \delta_1^*), \dots, (T_n^*, \delta_n^*)$ where $T_i^* = \min(Y_i^*, C_i^*)$ and $\delta_i^* = I(Y_i^* \leq C_i^*)$.

(The definition of $G_n(t)$ is obtained from that of $F_n(t)$ by replacing δ_i by $1 - \delta_i$.)

Efron's second proposal is to resample from the empirical distribution function of the observed minima and indicators.

Efron's bootstrap procedure II:

Resample $(T_1^*, \delta_1^*), \dots, (T_n^*, \delta_n^*)$ as a random sample with replacement from the pairs $\{(T_1, \delta_1), \dots, (T_n, \delta_n)\}$.

The first procedure could be called "model-based" since it uses the specific structure of random right censorship. The second procedure is more naive and could be called "model-free". An interesting result is that both procedures are identical if we assume that there are no ties between censored and uncensored values in the original sample. This means that both procedures give the same bootstrap values with the same probabilities.

The Kaplan-Meier estimator based on the bootstrapped observations

$(T_1^*, \delta_1^*), \dots, (T_n^*, \delta_n^*)$ is then given by

$$1 - F_n^*(t) = \prod_{T_i^* \leq t} \left(1 - \frac{m_i^*}{M_i^*}\right)^{\delta_i^*}$$

$$1 - F_n^*(t) = \prod_{T_i^* \leq t} \left(1 - \frac{m_i^*}{M_i^*}\right)^{\delta_i^*}$$

where $m_i^* = \#\{j : T_j^* = T_i^*\}$ and $M_i^* = \#\{j : T_j^* \geq T_i^*\}$. The bootstrapped Kaplan-Meier estimator can also be expressed in terms of the original observations $(T_1, \delta_1), \dots, (T_n, \delta_n)$. If F and G are continuous, we have

$$1 - F_n^*(t) = \prod_{T_{(i)} \leq t} \left(1 - \frac{M_{ni}}{\sum_{j=i}^n M_{nj}}\right)^{\delta_{(i)}} \quad (4.1)$$

where, for $i = 1, \dots, n$, M_{ni} = the number of times $(T_{(i)}, \delta_{(i)})$ appears in the bootstrap sample. We have that $(M_{n1}, \dots, M_{nn})$ has a multinomial $\left(n; \left(\frac{1}{n}, \dots, \frac{1}{n}\right)\right)$ distribution.

Throughout we will use the notations P^* , E^* , Var^* , ... for probability, expectation, variance, ... under the proposed resampling procedure (i.e. conditional on the original observations).

5. Weak convergence of the Kaplan-Meier process

Breslow and Crowley (1974) were the first to consider weak convergence of the process $n^{1/2}(F_n(t) - F(t))$, $0 \leq t \leq T$, where $T < T_H = \min(T_F, T_G)$. (Throughout we use the following notation: for any distribution function L , we write $T_L = \inf\{t : L(t) = 1\}$ for the right endpoint of the support of L .) Let $D[0, T]$ denote the space of right continuous functions with left hand limits.

THEOREM 1 (Breslow and Crowley). If $T < T_H$, then as $n \rightarrow \infty$,

$$n^{1/2}(F_n(\cdot) - F(\cdot)) \Rightarrow W(\cdot) \text{ in } D[0, T]$$

where $W(\cdot)$ is a Gaussian process with mean function 0 and covariance function

$$\Gamma_{st} = (1 - F(t))(1 - F(s)) \int_0^{\min(s, t)} \frac{1}{(1 - F(y))(1 - H(y-))} dF(y). \quad (5.1)$$

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- (1) Hall and Wellner (1980) observed that the limiting process $W(\cdot)$ has the following form in terms of a Brownian bridge process $B^o(\cdot)$ on $[0, 1]$:

$$W(t) = B^o(K(t)) \frac{1 - F(t)}{1 - K(t)} \quad (t < T_H)$$

$$\text{where } K(t) = \frac{C(t)}{1 + C(t)} \text{ and } C(t) = \int_0^t \frac{1}{(1 - F(y))(1 - H(y-))} dF(y).$$

(in case of no censoring: $K(t) = F(t)$ and $W(t) = B^o(F(t))$).

- (2) The result has been extended to weak convergence in $D[0, \infty)$ by Gill (1983).
- (3) From the result of Breslow and Crowley, it follows that, for each $t < T_H$,

$$n^{1/2}(F_n(t) - F(t)) \xrightarrow{d} N(0; \sigma^2(t))$$

where

$$\begin{aligned}\sigma^2(t) &= (1 - F(t))^2 \int_0^t \frac{1}{(1 - F(y))(1 - H(y-))} dF(y) \\ &= (1 - F(t))^2 \int_0^t \frac{1}{(1 - H(y))^2} dH^u(y) \quad (\text{if } F \text{ and } G \\ &\quad \text{are continuous}).\end{aligned}$$

- (4) The bootstrap can be used to estimate this asymptotic variance. The bootstrap estimator is given by

$$\hat{\sigma}_B^2(t) = n \text{Var}^*(F_n^*(t)).$$

If F and G are continuous, then we have from (4.1)

$$\ln(1 - F_n^*(t)) = \sum_{T_{(i)} \leq t} \delta_{(i)} \ln \left(1 - \frac{M_{ni}}{\sum_{j=1}^n M_{nj}} \right).$$

Since $(M_{n1}, \dots, M_{nn}) \sim \text{Multinomial}(n; (\frac{1}{n}, \dots, \frac{1}{n}))$, we have that the distribution of $\sum_{j=i}^n M_{nj}$ is Binomial $(n; \frac{n-i+1}{n})$ and also that the distribution of M_{ni} given that $\sum_{j=i}^n M_{nj} = s$ is Binomial $(s; \frac{1}{n-i+1})$. Using these facts, together with several "delta method" approximations, we obtain that $\hat{\sigma}_B^2(t) \approx \hat{\sigma}_G^2(t)$, where

$$\hat{\sigma}_G^2(t) = (1 - F_n(t))^2 n \sum_{T_{(i)} \leq t} \frac{\delta_{(i)}}{(n-i)(n-i+1)}$$

which is known as Greenwood's (1926) formula. Efron's bootstrap thus gives a justification for the use of this formula. In case of ties, the formula becomes

$$\hat{\sigma}_G^2(t) = (1 - F_n(t))^2 n \sum_{T_i \leq t} \frac{\delta_i}{M_i(M_i - m_i)} \quad (5.2)$$

- (5) The Greenwood estimator of the variance can be used to set up a large sample pointwise (at fixed time t) $100(1 - \alpha)\%$ confidence interval for $F(t)$. It is given by $F_n(t) \pm z_{\alpha/2} n^{-1/2} \hat{\sigma}_G(t)$, where $z_{\alpha/2} = \Phi^{-1}(1 - \frac{\alpha}{2})$ is the $(1 - \frac{\alpha}{2})$ quantile of the standard normal distribution function.

6. Almost sure asymptotic representations

In contrast to the classical empirical distribution function which is simply an average of i.i.d. Bernoulli distributed r.v., the Kaplan-Meier estimator does not have a simple structure. A very useful result of Lo and Singh (1986) represents the Kaplan-Meier estimator as an average of i.i.d. terms plus a lower order remainder term.

THEOREM 2 (Lo and Singh). Assume F continuous. For $t < T_H$:

$$F_n(t) = F(t) + \frac{1}{n} \sum_{i=1}^n \psi(T_i, \delta_i, t) + r_n(t) \quad (6.1)$$

where the $\psi(T_i, \delta_i, t)$ are i.i.d. zero mean r.v. which are bounded, uniformly in $0 \leq t \leq T$ (with $T < T_H$) and where

$$\sup_{0 \leq t \leq T} |r_n(t)| = O(n^{-3/4}(\log n)^{3/4}) \text{ a.s.}$$

REMARKS

- (1) The explicit expression for $\psi(T_i, \delta_i, t)$ is given by

$$\begin{aligned} \psi(T_i, \delta_i, t) = & (1 - F(t)) \left(\int_0^t \frac{I(T_i \leq y) - H(y)}{(1 - H(y))^2} dH^u(y) \right. \\ & + \frac{I(T_i \leq t, \delta_i = 1) - H^u(t)}{1 - H(t)} \\ & \left. - \int_0^t \frac{I(T_i \leq y, \delta_i = 1) - H^u(y)}{(1 - H(y))^2} dH(y) \right). \end{aligned}$$

- (2) We have that $\text{Cov}(\psi(T_i, \delta_i, t), \psi(T_i, \delta_i, s)) = \Gamma_{st}$ given by (5.1).
- (3) From a more general result of Major and Rejto (1988) it follows that the order bound of the remainder term can be sharpened to $O(n^{-1} \log n)$ a.s.
- (4) Immediate consequences of the representation theorem (theorem 2) are the asymptotic normality result for $n^{1/2}(F_n(t) - F(t))$, but also a law of the iterated logarithm result: $\sup_{0 \leq t \leq T} |F_n(t) - F(t)| = O(n^{-1/2}(\log \log n)^{-1/2})$ a.s. With some extra effort (checking tightness), also the weak convergence result of Breslow and Crowley (theorem 1) follows:

Lo and Singh (1986) also prove a bootstrap version of the above representation theorem.

THEOREM 3 (Lo and Singh). Assume F continuous. For $t < T_H$:

$$F_n^*(t) - F_n(t) = \frac{1}{n} \sum_{i=1}^n \psi(T_i^*, \delta_i^*, t) - \frac{1}{n} \sum_{i=1}^n \psi(T_i, \delta_i, t) + r_n^*(t)$$

with ψ as in theorem 2 and where for $T < T_H$:

$$\sup_{0 \leq t \leq T} |r_n^*(t)| = O_{P^*}(n^{-3/4}(\log n)^{3/4}) \quad \text{a.s.}$$

7. Weak convergence of the bootstrapped Kaplan-Meier process

The strong consistency of Efron's bootstrap for $n^{1/2}(F_n - F)$ has been established by entirely different techniques. For continuous F , weak convergence of $n^{1/2}(F_n^* - F_n)$ follows from theorem 3. For more general F , the result was obtained by Akritas (1986). His method uses point processes and martingale theory.

THEOREM 4 (Akritas). If $T < T_H$, then a.s. as $n \rightarrow \infty$,

$$n^{1/2}(F_n^*(\cdot) - F_n(\cdot)) \Rightarrow W(\cdot) \quad \text{in } D[0, T]$$

where $W(\cdot)$ is the same Gaussian process as for the original Kaplan-Meier process (see theorem 1).

As an application of the previous theorem, we mention the construction of large sample confidence bands for the distribution function F . The basic idea is the following: from theorem 1, we have that

$$\sup_{0 \leq t \leq T} n^{1/2} |F_n(t) - F(t)| \xrightarrow{d} \sup_{0 \leq t \leq T} |W(t)|.$$

Hence, if c_α is such that $P\left(\sup_{0 \leq t \leq T} |W(t)| \leq c_\alpha\right) = 1 - \alpha$, then the band $F_n(t) \pm c_\alpha n^{-1/2}$ ($0 \leq t \leq T$) has coverage probability approximately equal to $1 - \alpha$. By using theorem 3, c_α can be approximated by taking the $(1 - \alpha)$ quantile of the bootstrap distribution of $n^{1/2} \sup_{0 \leq t \leq T} |F_n^*(t) - F_n(t)|$. (See Akritas, 1986 for bootstrap versions of the Hall-Wellner, 1980 confidence bands.)

8. Other resampling plans

There exist other resampling methods for censored data, different from the one of Efron (1981) used above. Reid (1981) proposed (in a context of estimation of the median) to take an i.i.d. resample from the Kaplan-Meier estimator of F and to work with the corresponding empirical distribution function.

Reid's bootstrap procedure:

Resample $\tilde{T}_1, \dots, \tilde{T}_n \stackrel{i.i.d.}{\sim} F_n$.

It is clear that such a resample always consists entirely of originally uncensored T_i . If $\tilde{F}_n$ is the empirical distribution function of $\tilde{T}_1, \dots, \tilde{T}_n$, then Akritas (1986) showed that for $T < T_H$, a.s. as $n \rightarrow \infty$,

$$n^{1/2}(\tilde{F}_n(\cdot) - F_n(\cdot)) \Rightarrow \tilde{W}(\cdot) \text{ in } D[0, T]$$

but the Gaussian process $\tilde{W}$ does not agree with the limiting process W of the original Kaplan-Meier estimator. It follows that Reid's bootstrap cannot be used to approximate the quantile that is needed in the construction of the confidence band for F .

Hjort (1985) (in the context of Cox's regression model) proposed a resampling method as in Efron's Procedure I but with different construction of the censoring variables: if $\delta_i = 0$, then we observe the exact value of C_i and take $C_i^* = T_i$; if $\delta_i = 1$, then we know only that $C_i > T_i$ and we generate C_i^* from the Kaplan-Meier estimator of G , conditional on $C_i > T_i$.

Hjort's bootstrap procedure:

- (1) Resample $Y_1^*, \dots, Y_n^* \stackrel{i.i.d.}{\sim} F_n$.
- (2) Independently, resample $C_1^*, \dots, C_n^*$ as follows:
 - if $\delta_i = 0$, let $C_i^* = T_i$
 - if $\delta_i = 1$, generate C_i^* from G_{T_i} where

$$G_{T_i}(t) = (G_n(t) - G_n(T_i)) / (1 - G_n(T_i)) \quad (t \geq T_i).$$

- (3) Form $(T_1^*, \delta_1^*), \dots, (T_n^*, \delta_n^*)$ where $T_i^* = \min(Y_i^*, C_i^*)$ and $\delta_i^* = I(Y_i^* \leq C_i^*)$.

If F_n^* is the Kaplan-Meier estimator formed with the above (T_i^*, δ_i^*) , then Kim (1990) (see also Doss and Gill, 1992) showed that for $T < T_H$, a.s. as $n \rightarrow \infty$,

$$n^{1/2}(F_n^*(\cdot) - F_n(\cdot)) \Rightarrow W(\cdot) \text{ in } D[0, T]$$

where W is the same Gaussian process as for the original Kaplan-Meier process (see theorem 1).

9. Bootstrapping Kaplan-Meier quantiles

For $0 < p < 1$, we denote the p -th quantile of the lifetime distribution F by $\xi_p = F^{-1}(p) = \inf\{t : F(t) \geq p\}$. A simple estimator for ξ_p is the p -th quantile of the Kaplan-Meier estimator F_n : $\xi_{pn} = F_n^{-1}(p) = \inf\{t : F_n(t) \geq p\}$. Many large sample properties for quantile estimators can be derived from the asymptotic representations (Bahadur representations). They represent ξ_{pn} as follows:

$$\xi_{pn} = \xi_p + \frac{p - F_n(\xi_p)}{f(\xi_p)} + R_n(p) \quad (9.1)$$

where $f = F'$ and $R_n(p)$ is a remainder term. Cheng (1984) and Lo and Singh (1986) studied the a.s. behaviour of $R_n(p)$ in (9.1) and obtained the a.s. order $O(n^{-3/4}(\log n)^{3/4})$ under conditions on F'' , the second derivative of F . Under weaker conditions on F , Gijbels and Veraverbeke (1988) obtained (9.1) with $R_n(p) = o_P(n^{-1/2})$.

The main term $(p - F_n(\xi_p))/f(\xi_p)$ in (9.1) in its turn can be rewritten, by using representation (6.1), as an average of i.i.d. r.v. plus a remainder term. Weak convergence of the quantile process $n^{1/2}(\xi_{pn} - \xi_p)$ ($0 < p \leq p_0$) where $0 < p_0 < \min(1, T_{G(F^{-1})})$ was already obtained by Sander (1975).

The limiting process is then given by $-W(\xi_p)/f(\xi_p)$ ($0 < p \leq p_0$) where W is the limiting Gaussian process of the Kaplan-Meier process (see theorem 1).

Efron's bootstrap for Kaplan-Meier quantiles was studied in Lo and Singh (1986). Let F_n^* be the Kaplan-Meier estimator based on the bootstrapped observations $(T_1^*, \delta_1^*), \dots, (T_n^*, \delta_n^*)$ and let $\xi_{pn}^* = F_n^{*-1}(p) = \inf\{t : F_n^*(t) \geq p\}$. Their key result is a bootstrap version of the above representation theorem.

THEOREM 5 (Lo and Singh). Let $0 < p_1 \leq p_2 < \min(1, T_{G(F^{-1})})$. Suppose that F'' exists and is bounded on $[\xi_{p_0} - \eta, \xi_{p_1} + \eta]$ for some $\eta > 0$ and that $f = F'$ is bounded away from 0. Then,

$$\xi_{pn}^* - \xi_{pn} = -\frac{1}{n} \sum_{i=1}^n \frac{\psi(T_i^*, \delta_i^*, \xi_p)}{f(\xi_p)} + \frac{1}{n} \sum_{i=1}^n \frac{\psi(T_i, \delta_i, \xi_p)}{f(\xi_p)} + R_n^*(p)$$

with ψ as in theorem 2 and where $\sup_{p_1 \leq p \leq p_2} |R_n^*(p)| = O_p(n^{-3/4}(\log n)^{3/4})$ a.s.

This, and the application to confidence bands for the quantile function is also discussed in Doss and Gill (1992).

10. Bootstrapping in the submodel of proportional hazards

An important special case of the general right random censorship model is the Koziol-Green submodel of proportional hazards in which F and G are connected by the assumption that for some $\beta > 0$,

$$1 - G(t) = (1 - F(t))^\beta \quad (t > 0) \quad (10.1)$$

It is easily seen that, for continuous F , in this case $\gamma = E(\delta_1) = P(\delta_1 = 1) = \frac{1}{1+\beta}$ and that $1 - F(t) = (1 - H(t))^\gamma$.

Abdushukurov (1984) and Cheng and Lin (1984, 1987) independently proposed to estimate $F(t)$ by $\tilde{F}_n(t)$ (now called the ACL estimator), given by

$$\tilde{F}_n(t) = 1 - (1 - H_n(t))^{\gamma_n}$$

where $H_n(t) = \frac{1}{n} \sum_{i=1}^n I(T_i \leq t)$ and $\gamma_n = \frac{1}{n} \sum_{i=1}^n \delta_i$.

An important characterization of the proportional hazards model is that (10.1) holds for some $\beta > 0$ if and only if T_1 and δ_1 are independent. (See for example Sethuraman, 1965.) Therefore, a resampling scheme which mimics the model is the following one, proposed by Dikta and Ghorai (1990): resample independently $T_1^*, \dots, T_n^*$ from H_n and $\delta_1^*, \dots, \delta_n^*$ from a Bernoulli distribution with parameter γ_n . Then let $H_n^*(t) = \frac{1}{n} \sum_{i=1}^n I(T_i^* \leq t)$ and $\gamma_n^* = \frac{1}{n} \sum_{i=1}^n \delta_i^*$ and use as a bootstrap estimator for $\tilde{F}_n(t)$:

$$\tilde{F}_n^*(t) = 1 - (1 - H_n^*(t))^{\gamma_n^*}.$$

Dikta and Ghorai (1990) proved that for F continuous and for $T < T_H$, a.s. as $n \rightarrow \infty$,

$$n^{1/2}(\tilde{F}_n^*(\cdot) - \tilde{F}_n(\cdot)) \Rightarrow V(\cdot) \text{ in } D[0, T]$$

where V is the same process that appears as the limit in $D[0, T]$ of $n^{1/2}(\tilde{F}_n(\cdot) - F(\cdot))$ (see Cheng and Lin, 1987). The process V is Gaussian with zero mean and covariance function given by $(0 \leq s \leq t \leq T)$:

$$\begin{aligned} \text{Cov}(V(t), V(s)) = & (1 - H(s))(1 - H(t)) \left\{ \gamma^2 \frac{H(s)}{1 - H(s)} \right. \\ & \left. + \gamma(1 - \gamma) \log(1 - H(s)) \log(1 - H(t)) \right\} \end{aligned}$$

Estimation of quantiles of F under this model and their bootstrapping has been studied in Gijbels and Veraverbeke (1989) and Veraverbeke (1994).

11. Accuracy of the bootstrap with censored data (second order asymptotics)

In section 7 we have seen that the bootstrap distribution estimator is consistent: that is the bootstrap error tends to zero a.s. The question arises for finding the rate of this convergence so that the bootstrap approximation can be compared to other approximations using a one-term Edgeworth expansion.

For uncensored data it is well known that for some classes of statistics and under regularity conditions, the bootstrap approximation is more accurate than the normal approximation. While the error of the normal approximation is typically $O(n^{-1/2})$ (by a Berry-Esseen theorem), the rate of convergence of the bootstrap distribution estimator can, in some situations, be shown to be $o(n^{-1/2})$ a.s. (which is typically that of a one-term Edgeworth expansion).

For sample means, such results have been obtained by Bickel and Freedman (1981), Singh (1981), Abramovitch and Singh (1985) and many others. More general than sample means, Helmers (1991) considered the case of a U -statistic

$$U_n = \binom{n}{2}^{-1} \sum_{1 \leq i \leq j \leq n} h(X_i, X_j)$$

with symmetric kernel h and based on an i.i.d. sample $X_1, \dots, X_n$. Let $g(X_1) = E[h(X_1, X_2)|X_1] - E[h(X_1, X_2)]$ and let $\hat{\sigma}_n^2$ be the jackknife estimator for $n \text{Var}(U_n)$. Then Helmers (1991) showed that the convergence rate of the bootstrap estimator of the distribution of the Studentized statistic $n^{1/2}(U_n - EU_n)/\hat{\sigma}_n$ is $o(n^{-1/2})$ a.s., provided $E|h(X_1, X_2)|^{4+\varepsilon} < \infty$, $E|h(X_1, X_1)|^3 < \infty$ and the distribution of $g(X_1)$ is non-lattice. (Helmers' moment conditions can be somewhat relaxed as shown by Putter (1994)). The proof of such results is typically very technical and relies on Edgeworth expansions for the Studentized statistics and for the corresponding bootstrap versions.

As will be seen below (theorems 6, 7, 9 and 10) the results for U -statistics are basic for deriving second-order results for more complicated statistics. See also Babu (1991). For a survey on bootstrapping U -statistics, we refer to Janssen (1997).

Chen and Lo (1996) derived the Edgeworth expansions for the Studentized Kaplan-Meier estimator $n^{1/2}(F_n(t) - F(t))/\hat{\sigma}_G(t)$ and for the bootstrapped version. Here $\hat{\sigma}_G^2(t)$ is given by the Greenwood formula (5.2). The one-term Edgeworth expansion is given in the next theorem.

THEOREM 6 (Chen and Lo). Assume F and G continuous. Let $t > 0$ be such that $1 - H(t) > 0$ and $G(t) > 0$. Then, as $n \rightarrow \infty$,

$$\sup_{x \in \mathbb{R}} |P(n^{1/2}(F_n(t) - F(t))/\hat{\sigma}_G(t) \leq x) - \{\Phi(x) + n^{-1/2}\varphi(x)(ax^2 + b)\}| \\ = o(n^{-1/2})$$

where Φ is the standard normal distribution function, $\varphi = \Phi'$, and a and b are constants depending on F, G and t only.

The proof relies on a representation whose main term is a U -statistic and to which the Edgeworth expansion result of Bickel, Götze and van Zwet (1986) can be applied. Edgeworth expansions typically require a non-lattice condition and moment conditions. In the present situation, the condition in theorem 6 is strong enough to guarantee that the kernel of the U -statistic is bounded and has a continuous distribution.

The next theorem gives the one-term Edgeworth expansion for the bootstrapped Studentized Kaplan-Meier estimator. Recall that the bootstrap sample $(T_1^*, \delta_1^*), \dots, (T_n^*, \delta_n^*)$ is obtained by taking a sample with replacement from the set of observations $\{(T_1, \delta_1), \dots, (T_n, \delta_n)\}$. In establishing a one-term Edgeworth expansion for $P^*(n^{1/2}(F_n^*(t) - F_n(t))/\hat{\sigma}_G^*(t) \leq x)$ it is not possible to use the same proof as in theorem 6, since this one relies on the continuity of F and G in the assumption. To avoid ties in the bootstrap sample a slightly smoothened version of its empirical distribution function has to be considered. A smoothed bootstrap sample $(T_1', \delta_1'), \dots, (T_n', \delta_n')$ is taken from the density function

$$\frac{1}{2n\varepsilon_n} \sum_{i=1}^n I(|t' - T_i| \leq \varepsilon_n, \delta' = \delta_i)$$

which, for some appropriately chosen small $\varepsilon_n > 0$ has a distribution function which is close to the usual empirical distribution function. Defining $T_i^* = T_j$ if $|T_i' - T_j| \leq \varepsilon_n$ for some $1 \leq j \leq n$, we have that $(T_1^*, \delta_1^*), \dots, (T_n^*, \delta_n^*)$ is the ordinary bootstrap sample.

THEOREM 7 (Chen and Lo). Assume F and G continuous. Let $t > 0$ be such that $1 - H(t) > 0$ and $G(t) > 0$. Then a.s. as $n \rightarrow \infty$,

$$\sup_x |P^*(n^{1/2}(F_n^*(t) - F_n(t))/\hat{\sigma}_G^*(t) \leq x) - \{\Phi(x) + n^{-1/2}\varphi(x)(ax^2 + b)\}| = o(n^{-1/2}).$$

An important corollary from theorems 6 and 7 is that under the given assumption, a.s. as $n \rightarrow \infty$,

$$\sup_x |P^*(n^{1/2}(F_n^*(t) - F_n(t))/\hat{\sigma}_G^*(t) \leq x) - P(n^{1/2}(F_n(t) - F(t))/\hat{\sigma}_G(t) \leq x)| = o(n^{-1/2}).$$

This shows that bootstrapping the Studentized Kaplan-Meier estimator offers a better alternative than the normal approximation. (Compare with the $O(n^{-1/2})$ rate of the Berry-Esseen theorem in Chang and Rao, 1989, Janssen and Veraverbeke 1992.)

REMARK. A completely similar set of results has recently been obtained by Wang and Zheng (1995). They studentize the Kaplan-Meier in a different way by considering $n^{1/2}(F_n(t) - F(t))/\hat{\sigma}_J(t)$ where $\hat{\sigma}_J^2(t)$ is the jackknife estimator for $\sigma^2(t)$, that is

$$\hat{\sigma}_J^2(t) = \frac{1}{n} \sum_{i=1}^n (J_{ni}(t) - \bar{J}_n(t))^2$$

where

$$J_{ni}(t) = nF_n(t) - (n-1)F_{n-1}^{(-i)}(t), (i = 1, \dots, n), \bar{J}_n(t) = \frac{1}{n} \sum_{i=1}^n J_{ni}(t),$$

$F_{n-1}^{(-i)}(t)$ is the Kaplan-Meier estimator based on $(T_1, \delta_1), \dots, (T_{i-1}, \delta_{i-1}), (T_{i+1}, \delta_{i+1}), \dots, (T_n, \delta_n)$ (see also Singh and Liu, 1990). Another difference is that Wang and Zheng use Efron's smoothed bootstrap: the Y_i^* and C_i^* are independently resampled from F_{nh} and G_{nh} respectively, where $F_{nh}(t) = \int_0^\infty K_o((t-s)/h_n) dF_n(s)$, $G_{nh}(t) = \int_0^\infty K_o((t-s)/h_n) dG_n(s)$ are kernel smoothed versions of the original Kaplan-Meier estimators F_n and G_n , h_n is a sequence of positive constants tending to zero and K_o is a probability distribution function.

12. Regression models with randomly censored observations

Very often it happens that together with the observation of the i -th individual's lifetime Y_i or censoring time C_i , one also has information on other characteristics X_i (covariates, explanatory variables, design variables). For example: dose of a drug, blood pressure of a patient, ... Regression models study the effect of these covariates on the distribution of the true lifetimes. The problem is to estimate the conditional distribution function

$$F_x(t) = P(Y \leq t \mid X = x)$$

or functionals of this, such as the mean regression function $m(x) = E(Y \mid X = x) = \int_0^\infty t dF_x(t)$ or regression quantiles $F_x^{-1}(p) = \inf\{t : F_x(t) \geq p\}$.

Regression analysis with randomly censored data has been studied by many authors. Widely studied is the proportional hazards model of Cox (1972, 1975) which specifies the structure of the hazard rate function and where parameters

are estimated by the partial likelihood method. Alternative approaches extend the complete data least squares estimation to the censored data case: Miller (1976), Buckley and James (1979), Koul, Susarla and Van Ryzin (1981), Leurgans (1987), Recently there have been a number of contributions that do not require restrictive assumptions and are easy computationally. For example, estimators based on monotone estimating equations have been proposed by Fygenson and Ritov (1994) and Akritas, Murphy and LaValley (1995). Also, in Akritas (1994) and Akritas (1996), nonparametric regression techniques are used to construct versions of the least squares estimator. These papers give detailed references to the other approaches and lend themselves quite readily to bootstrapping since they are based on nonparametric regression methods.

There are also nonparametric methods for fitting a completely unspecified regression function. Fan and Gijbels (1994, 1996) proposed a method based on transforming the data, followed by a local linear regression. Kernel estimators for the conditional distribution function $F_x(t)$ were first introduced by Beran (1981) and further studied by Doksum and Yandell (1982), Horváth (1981), Dabrowska (1987, 1989), González-Manteiga and Cadarso-Suarez (1994), McKeague and Utikal (1990),

13. The proportional hazards regression model of Cox

In the proportional hazards regression model of Cox (1972), the relation between a, possibly right censored, lifetime Y and the covariate X is modelled via the hazard rate function

$$\lambda_x(t) = \lim_{h \rightarrow 0} P(Y \leq t+h | Y > t; X=x)$$

(For simplicity we assume here that the covariate X is one-dimensional, but generalizations are possible.) Cox's proportional hazards model specifies that $\lambda_x(t)$ is given by

$$\lambda_x(t) = \lambda_0(t)e^{\beta x} \quad (13.1)$$

where $\lambda_0(t)$ is an unspecified baseline hazard rate function (the hazard for an individual with $x=0$) and β is an unknown regression parameter.

Suppose that there are n individuals in the study and that the observations are given as $(X_1, T_1, \delta_1), \dots, (X_n, T_n, \delta_n)$ where for individual i , X_i denotes the covariate and where, as before, $T_i = \min(Y_i, C_i)$ and $\delta_i = I(Y_i \leq C_i)$.

Cox's maximum partial likelihood estimator for β (Cox, 1975) is the value for β maximizing the partial likelihood function

$$\tilde{L}_n(\beta) = \prod_{i=1}^n \left(\frac{e^{\beta X_i}}{\sum_{j=1}^n e^{\beta X_j} I(T_j \geq T_i)} \right)^{\delta_i}$$

(or the log partial likelihood function $\log \tilde{L}_n(\beta)$).

The large sample properties of Cox's maximum partial likelihood estimator were established by Tsiatis (1981) (see also Andersen and Gill, 1982, for a counting process approach). He considered a slightly modified version of $\tilde{L}_n$:

$$L_n(\beta) = \prod_{\substack{i=1 \\ T_i \leq T}}^n \left(\frac{e^{\beta X_i}}{\sum_{j=1}^n e^{\beta X_j} I(T_j \geq T_i)} \right)^{\delta_i}$$

where $T < \infty$ is some prespecified time satisfying $P(T_i \geq T) > 0$. Since $L_n(\beta)$ is differentiable, a maximum partial likelihood function solves the equation

$$U_n(\beta) = 0 \quad (13.2)$$

where $U_n(\beta) = \frac{\partial}{\partial \beta} \log L(\beta)$. Tsiatis (1981) proved strong consistency and asymptotic normality.

THEOREM 8 (Tsiatis). Assume

- (i) The covariate X is a r.v. with density f .
- (ii) The random vectors $(X_1, Y_1, C_1), \dots, (X_n, Y_n, C_n)$ are i.i.d.
- (iii) Y_i and C_i are independent, given the covariate X_i .
- (iv) $E[(Xe^{\beta X})^2]$ is bounded, uniformly in a neighbourhood of β .

Then, there is a solution $\hat{\beta}_n$ of equation (13.2) such that

1. $\hat{\beta}_n \rightarrow \beta$ a.s.

2. $n^{1/2}(\hat{\beta}_n - \beta) \xrightarrow{d} N(0; \sigma^2(\beta))$ where $\sigma^2(\beta) = \frac{1}{I(\beta)}$ with

$$I(\beta) = \int_0^T \left[\frac{\alpha_2(t, \beta)}{\alpha_0(t, \beta)} - \left(\frac{\alpha_1(t, \beta)}{\alpha_0(t, \beta)} \right)^2 \right] \alpha_0(t) \lambda_0(t) dt$$

$$\alpha_k(t, \beta) = \int x^k e^{\beta x} P(T \geq t | X = x) f(x) dx \quad k = 0, 1, 2.$$

The asymptotic variance $\sigma^2(\beta) = \frac{1}{I(\beta)}$ can be consistently estimated by $\frac{n}{I_n(\hat{\beta}_n)}$ where $I_n(\beta) = -\frac{\partial}{\partial \beta} U_n(\beta)$ is the information function. Hence the Studentized estimator is $(I_n(\hat{\beta}_n))^{1/2}(\hat{\beta}_n - \beta)$.

14. Bootstrapping Cox's proportional hazards model

As suggested by Efron and Tibshirani (1986), an obvious bootstrap procedure for Cox's proportional hazards regression model is to resample from the triples, then calculate the partial likelihood and then maximize it.

Efron and Tibshirani's bootstrap procedure:

- (1) Resample $(X_1^*, T_1^*, \delta_1^*), \dots, (X_n^*, T_n^*, \delta_n^*)$ i.i.d. from the empirical distribution function of $(X_1, T_1, \delta_1), \dots, (X_n, T_n, \delta_n)$.
- (2) Calculate $\hat{\beta}_n^*$: value of β such that

$$L_n^*(\beta) = \prod_{\substack{i=1 \\ T_i^* \leq T}}^n \left(\frac{e^{\beta X_i^*}}{\sum_{j=1}^n e^{\beta X_j^*} I(T_j^* \geq T_i^*)} \right)^{\delta_i^*}$$

is maximized.

Also define $U_n^*(\beta) = \frac{\partial}{\partial \beta} \log L_n^*(\beta)$, $I_n^*(\beta) = -\frac{\partial}{\partial \beta} U_n^*(\beta)$.

Gu (1992) established the one-term Edgeworth expansions for $n^{1/2}(\hat{\beta}_n - \beta)$ and related quantities. He also showed that the bootstrap approximation is second order correct. The method is again by U -statistic approximation and application of the Edgeworth expansion result for U -statistics. The assumptions in Gu (1992) are:

- (i) X is bounded uniformly.
- (ii) The range of β is bounded.
- (iii) The (X_i, Y_i, C_i) are i.i.d. and, given $X_i : Y_i$ and C_i are independent.
- (iv) λ_0 in (13.1) is continuous.
- (v) $I(\beta) > 0$.

Under these conditions he established one-term Edgeworth expansions for the following (asymptotically $N(0, 1)$) quantities:

$$\sigma(\beta) \frac{U_n(\beta)}{\sqrt{n}}, \frac{U_n(\beta)}{\sqrt{I_n(\beta)}}, \frac{U_n(\beta)}{\sqrt{I_n(\hat{\beta}_n)}}, n^{1/2} \frac{(\hat{\beta}_n - \beta)}{\sigma(\beta)}, \sqrt{I_n(\hat{\beta}_n)} (\hat{\beta}_n - \beta).$$

For example, for the Studentized statistic, we have:

THEOREM 9 (Gu). Assume (i) – (v) above. Then, as $n \rightarrow \infty$,

$$\sup_{x \in \mathbb{R}} |P(\sqrt{I_n(\hat{\beta}_n)} (\hat{\beta}_n - \beta) \leq x) - \{\Phi(x) + n^{-1/2} \varphi(x)(c_1 x^2 + c_2)\}| = o(n^{-1/2})$$

for some constants c_1 and c_2 .

The constants c_1 and c_2 can be replaced by consistent estimators and the results remain true.

Another way of obtaining good estimates is by the bootstrap method. It is shown that, under the same assumptions, the bootstrap distributions are good approximations for the sampling distributions of the five quantities. We state only one example.

THEOREM 10 (Gu). Assume (i) – (v) above. Then, a.s. as $n \rightarrow \infty$,

$$\sup_{x \in \mathbb{R}} |P^*(\sqrt{I_n^*(\hat{\beta}_n^*)} (\hat{\beta}_n^* - \hat{\beta}_n) \leq x) - P(\sqrt{I_n(\hat{\beta}_n)} (\hat{\beta}_n - \beta) \leq x)| = o(n^{-1/2}).$$

This shows that the bootstrap distribution is more accurate than the normal approximation.

15. Nonparametric regression

The relation between the distribution of the lifetime Y and the value of the covariate X has also been analyzed in a completely nonparametric way, that is, without assuming any condition on the regression function. Beran (1981) extended the definition of the Kaplan-Meier estimator to the regression context by proposing an entirely nonparametric estimator for $F_x(t)$ based on a random sample of observations $(X_1, T_1, \delta_1), \dots, (X_n, T_n, \delta_n)$ (with as before: $T_i = \min(Y_i, C_i)$ and $\delta_i = I(Y_i \leq C_i)$). It is assumed that the r.v. Y_i and C_i are conditionally independent, given X_i . This ensures identifiability of $F_x(t)$ and the corresponding cumulative hazard function

$$\Lambda_x(t) = \int_0^t \frac{1}{1 - F_x(s-)} dF_x(s).$$

More precisely we have

$$\Lambda_x(t) = \int_0^t \frac{1}{1 - H_x(s-)} dH_x^u(s) \quad (15.1)$$

where $H_x(t) = P(T_1 \leq t \mid X = x)$ and $H_x^u(t) = P(T_1 \leq t, \delta_1 = 1 \mid X = x)$.

And also

$$1 - F_x(t) = \exp(-\Lambda_x^c(t)) \prod_{s \leq t} (1 - \Delta \Lambda_x(s))$$

where $\Delta \Lambda_x(t) = \Lambda_x(t) - \Lambda_x(t-)$ and $\Lambda_x^c(t) = \Lambda_x(t) - \sum_{s \leq t} \Delta \Lambda_x(s)$.

Beran suggests to replace $H_x(t)$ and $H_x^u(t)$ in (15.1) by estimators of the type

$$H_{xn}(t) = \sum_{i=1}^n W_{ni}(x) I(T_i \leq t)$$

$$H_{xn}^u(t) = \sum_{i=1}^n W_{ni}(x) I(T_i \leq t, \delta_i = 1)$$

respectively, where $\{W_{ni}(x)\}$, $i = 1, \dots, n$ is some set of probability weights depending on the covariates $X_1, \dots, X_n$ (for instance kernel type weights or nearest neighbour type weights). This then gives an estimator for $\Lambda_x(t)$

$$\Lambda_{xn}(t) = \int_0^t \frac{1}{1 - H_{xn}(s-)} dH_{xn}^u(s)$$

and Beran's estimator for $F_x(t)$ is then defined as

$$1 - F_{xn}(t) = \prod_{s \leq t} (1 - \Delta \Lambda_{xn}(s)).$$

This estimator has been studied by several people. Beran (1981) proved uniform consistency. Dabrowska (1987, 1989, 1992) obtained weak convergence results and also studied the corresponding quantile functions. McKeague and Utikal (1990) provided the counting process treatment and extended the regression model. González-Manteiga and Cadarso-Suarez (1994) and Van Keilegom and Veraverbeke (1997b) proved a.s. asymptotic representations in the fixed design case (i.e. nonrandom covariates X_i). In Van Keilegom and Veraverbeke (1996, 1997a, 1997c), a further study was made of this Beran extension of the Kaplan-Meier estimator, the corresponding quantile estimator and the bootstrap versions.

16. Bootstrapping in the fixed design regression model

Let us consider Beran's estimator for the situation of fixed design points $0 \leq x_1 \leq \dots \leq x_n \leq 1$. As in the uncensored case, it is then natural to use Gasser-Müller type weights in the construction of Beran's estimator. They are given by

$$w_{ni}(x; h_n) = \frac{1}{c_n(x; h_n)} \int_{x_{i-1}}^{x_i} \frac{1}{h_n} K\left(\frac{x-z}{h_n}\right) dz \quad i = 1, \dots, n$$

$$c_n(x; h_n) = \int_0^{x_m} \frac{1}{h_n} K\left(\frac{x-z}{h_n}\right) dz.$$

Here $x_0 = 0$, K is a known probability density function (kernel) and $\{h_n\}$ is a sequence of positive constants (bandwidth) tending to zero as $n \rightarrow \infty$. It is easily seen that Beran's estimator is given by

$$F_{xh}(t) = 1 - \left\{ \prod_{T_{(i)} \leq t} \left(1 - \frac{w_{n(i)}(x; h_n)}{1 - \sum_{j=1}^{i-1} w_{n(j)}(x; h_n)} \right)^{\delta_{(i)}} \right\} I(t < T_{(n)}). \quad (16.1)$$

Here $T_{(1)} \leq \dots \leq T_{(n)}$ are the ordered $T_1, \dots, T_n$ and the $\delta_{(i)}$ and $w_{n(i)}(x; h_n)$ are the corresponding δ_i and $w_{ni}(x; h_n)$. In case of ties, we make the usual convention that uncensored observations are considered to occur just before censored observations. Clearly $F_{xh}(t)$ is a step function with

jumps only at the uncensored observations. Note that if we think all weights $w_{ni}(x; h_n)$ equal to n^{-1} , then $F_{xh}(t)$ becomes the classical Kaplan-Meier estimator. On the other hand, in the absence of censoring ($T_i = Y_i$ and $\delta_i = 1$ for all i) the estimator equals $\sum_{i=1}^n w_{ni}(x; h_n) I(Y_i \leq t)$, which is the kernel estimator of Stone (1977).

The results require the following regularity conditions:

- C1 $x_n \rightarrow 1$, $\Delta_n = \max_{1 \leq i \leq n} (x_i - x_{i-1}) = O(n^{-1})$
 $n(\Delta_n - \delta_n) = o(1)$ where $\delta_n = \min_{1 \leq i \leq n} (x_i - x_{i-1})$.
- C2 K is a probability density function with finite support $[-L, L]$ for some $L > 0$, $\int uK(u)du = 0$ and k is Lipschitz of order 1.
- C2' K is a twice differentiable probability density function with finite support $[-L, L]$ for some $L > 0$, $\int uK(u)du = 0$, K'' is continuous, and $K(-L) = K'(-L) = K(L) = K'(L) = 0$.
- C3 $H_x(t)$ and $H_x^u(t)$ have continuous first and second order partial derivatives with respect to x and t in $(x, t) \in [0, 1] \times [0, T]$, where $T < T_{H_x} = \min(T_{F_x}, T_{G_x})$.

In Van Keilegom and Veraverbeke (1997b) the following representation theorem was proved. It generalizes the result of Lo and Singh (1986) (see theorem 2) to the regression case.

THEOREM II. Assume (C1) (C2) (C3), $h_n \rightarrow 0$, $\frac{\log n}{nh_n} \rightarrow 0$, $\frac{nh_n^5}{\log n} = O(1)$.

Then, for $t < T_{H_x}$:

$$F_{xh}(t) = F_x(t) + \sum_{i=1}^n w_{ni}(x; h_n) g_{tx}(T_i, \delta_i) + r_n(t, x)$$

where the $g_{tx}(T_i, \delta_i)$ are i.i.d. r.v. and where, for $T < T_{H_x}$

$$\sup_{0 \leq t \leq T} |r_n(t, x)| = O((nh_n)^{-3/4} (\log n)^{3/4}) \quad \text{a.s.}$$

From this the asymptotic normality result follows and with some extra effort (see Van Keilegom and Veraverbeke (1997a)) the weak convergence result below can be obtained (see theorem 12).

We use the notation $W_x(\cdot)$ for a Gaussian process with mean function zero and covariance function

$$\Gamma_{st} = \|K\|_2^2 (1 - F_x(t))(1 - F_x(s)) \int_0^{\min(s, t)} \frac{1}{(1 - H_x(y))^2} dH_x^u(y)$$

and the notation $\tilde{W}_x(\cdot)$ for a Gaussian process with mean function $b_x(t)$ and covariance function Γ_{st} , where

$$b_x(t) = \frac{1}{2}(1 - F_x(t)) \int_0^t \left\{ \frac{\ddot{H}_x(s) dH_x^u(s)}{(1 - H_x(s))^2} + \frac{d\ddot{H}_x^u(s)}{1 - H_x(s)} \right\} \mu_2^K C^{5/2}.$$

Here $\ddot{H}_x = \frac{\partial^2}{\partial x^2} H_x$, $\ddot{H}_x^u = \frac{\partial^2}{\partial x^2} H_x^u$ and $\mu_2^K = \int u^2 K(u) du$.

THEOREM 12. Assume (C1) (C2) (C3). Let $T < T_{H_x}$.

(a) If $nh_n^5 \rightarrow 0$ and $\frac{(\log n)^3}{nh_n} \rightarrow 0$, then as $n \rightarrow \infty$,

$$(nh_n)^{1/2} (F_{xh}(\cdot) - F_x(\cdot)) \Rightarrow W_x(\cdot) \quad \text{in } D[0, T].$$

(b) If $h_n = Cn^{-1/5}$ for some $C > 0$, then as $n \rightarrow \infty$,

$$(nh_n)^{1/2} (F_{xh}(\cdot) - F_x(\cdot)) \rightarrow \tilde{W}_x(\cdot) \quad \text{in } D[0, T].$$

Note that the (b)-part covers the "optimal bandwidth" case, that is the bandwidth which minimizes the approximate mean squared error.

In Van Keilegom and Veraverbeke (1997b), the following resampling scheme was proposed to construct a bootstrap version of $F_{xh}(t)$. The procedure makes use of estimators $F_{x_i g}$ and $G_{x_i g}$. Here $F_{x_i g}$ is an estimator for F_{x_i} as in (16.1), but with a bandwidth sequence $\{g_n\}$ which is different from $\{h_n\}$. The distribution $G_{x_i g}$ is the analogous estimator for G_{x_i} . The bandwidth sequence $\{g_n\}$ is typically such that $g_n/h_n \rightarrow \infty$ as $n \rightarrow \infty$ ("oversmoothing").

Bootstrap procedure:

- (1) $Y_1^*, \dots, Y_n^*$ are independent and $Y_i^* \sim F_{x_i g}$
- (2) Independently: $C_1^*, \dots, C_n^*$ are independent and $C_i^* \sim G_{x_i g}$
- (3) Form $T_i^* = \min(Y_i^*, C_i^*)$ and $\delta_i^* = I(Y_i^* \leq C_i^*)$.

It is easily verified that the above procedure is equivalent to the one where the pairs (T_i^*, δ_i^*) are drawn (with replacement) from $(T_1, \delta_1), \dots, (T_n, \delta_n)$, giving probability $w_{nj}(x_i; h_n)$ to (T_j, δ_j) for $j=1, \dots, n$. Based on the bootstrap sample $(T_1^*, \delta_1^*), \dots, (T_n^*, \delta_n^*)$ the bootstrap analogue of the Kaplan-Meier type estimator in (16.1) is given by

$$F_{xhg}^*(t) = 1 - \left\{ \prod_{T_{(i)}^* \leq t} \left(1 - \frac{w_{n(i)}(x; h_n)}{1 - \sum_{j=1}^{i-1} w_{n(j)}(x; h_n)} \right)^{\delta_{(i)}^*} \right\} I(t < T_{(n)}^*)$$

where $T_{(1)}^* \leq \dots \leq T_{(n)}^*$ and $\delta_{(i)}^*$ and $w_{n(i)}(x; h_n)$ correspond to $T_{(i)}^*$.

In Van Keilegom and Veraverbeke (1997a) the following strong consistency of the bootstrap is shown.

THEOREM 13. Assume (C1) (C2') (C3). Let $T < T_{H_x}$, $h_n = Cn^{-1/5}$ for some $C > 0$, $g_n \rightarrow 0$, $\frac{ng_n^5}{\log n} \rightarrow \infty$, $\frac{ng_n^5}{\log n} \cdot \frac{h_n}{g_n} = O(1)$.

Then, a.s. as $n \rightarrow \infty$,

$$(nh_n)^{1/2}(F_{xhg}^*(\cdot) - F_{xg}(\cdot)) \Rightarrow \tilde{W}_x(\cdot) \text{ in } D[0, T]$$

where $\tilde{W}_x(\cdot)$ is the same process as in theorem 12.

Similar results are obtained for the corresponding bootstrapped quantile process $(nh_n)^{1/2}(F_{xhg}^{*-1}(\cdot) - F_{xg}^{-1}(\cdot))$.

17. References

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